

VEIN: On-Chain Causal Risk Attribution for Cryptocurrency Systemic Risk

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Abstract

Existing causal measures of systemic risk in cryptocurrency markets infer the contagion graph *statistically* from price returns, using constraint-based structure learning or directed-information estimation. We propose VEIN (Verifiable Entity-resolved Interventional/counterfactual Network risk), a framework that instead treats the *observed* on-chain transaction graph as the structural backbone of a Pearl-style structural causal model (SCM): the causal parent set of each economic entity is read off the ledger rather than estimated from correlations. On this graph VEIN defines an interventional risk measure (On-Chain Causal CoVaR) through the do-operator and, to our knowledge, the first counterfactual (Pearl Level-3) systemic-risk attribution measure in finance, which decomposes a realised loss into the portion attributable to a specific upstream entity’s distress. We implement the complete pipeline—a four-tier entity-resolution layer over Dune Analytics’ decoded Ethereum tables, the SCM, and the risk measures—and evaluate it on real on-chain data around the October 2025 cascading liquidation event, the largest deleveraging episode in crypto history. At the event’s native (hourly) timescale, the observed graph matches or slightly outperforms a price-estimated Granger graph at predicting the cascade; the OC-CoVaR systemic ranking diverges from Δ CoVaR and MES (Spearman $\rho = \mathbf{0.52}$); the counterfactual measure attributes concrete loss shares between exchanges; and a battery of causal-validity checks (placebo/negative controls, unobserved-confounding robustness values, resolution robustness, and external validity against the documented event narrative) corroborates the framework. The one assumption that does *not* survive falsification is that flow *direction* encodes causal direction: edge reversal barely changes predictions, indicating that causal information resides in the network’s connectivity structure rather than in flow orientation. We discuss

the implications for regulators and the limits imposed by centralised-exchange opacity. JEL classification: G01, G17, G18, C58.

Keywords: systemic risk, cryptocurrency, structural causal models, counterfactual inference, blockchain, CoVaR, contagion

1 Introduction

On 10–11 October 2025, cryptocurrency markets experienced the largest deleveraging event in their history: on the order of nineteen billion US dollars of leveraged positions were liquidated within thirty-six hours.¹ Early scholarly post-mortems of the episode [1, 2] document a striking feature that sets it apart from the 2022 episodes (Terra/Luna, Celsius, FTX): whereas those were rooted in fundamental insolvency, the October 2025 cascade was largely *mechanical*. A mispricing in a centralised exchange’s internal valuation engine² propagated into automated liquidations, which in turn forced sales that depressed prices and triggered further liquidations—a classic funding-liquidity spiral operating at machine speed. A defining feature was architectural: decentralised finance (DeFi) lending protocols with transparent, over-collateralised positions (e.g. Aave) emerged with essentially zero bad debt, whereas centralised-exchange (CEX) risk engines amplified the shock. The synchronised depeg of the synthetic dollar USDe—which remained fully solvent throughout, so that the depeg was a pricing and liquidity artefact rather than a default—is emblematic of this architecture-dependent fragility.³ Crucially for risk measurement, the proximate

¹For event overviews and the minute-level liquidation mechanics, see CoinDesk Research, <https://www.coindesk.com/research/market-spotlight-the-19-billion-liquidation-that-shook-crypto>; CoinGecko, <https://www.coingecko.com/learn/october-10-crypto-crash-explained>; and FTI Consulting, <https://www.fticonsulting.com/insights/articles/crypto-crash-october-2025-leverage-met-liquidity>.

²The proximate trigger has been attributed to Binance’s unified-account margin system, which valued posted collateral (including USDe, wBETH and BNSOL) using its own internal spot prices rather than external oracles; see <https://www.ccn.com/education/crypto/ethena-usde-depeg-binance-crash-explained/> and <https://ambcrypto.com/not-a-true-depeg-says-ethena-founder-as-usde-faces-binance-only-slip/>.

³USDe’s price fell to roughly \$0.65 on Binance while holding near \$1 on other venues and in DeFi, and remained redeemable as supply contracted from about \$9bn to \$6bn, confirming an exchange-specific pricing artefact rather than insolvency; see CoinDesk, <https://www.coindesk.com/markets/2025/10/11/ethena-s-usde-briefly-loses-peg-during-usd19b-crypto-liquidation-cascade> and <https://www.coindesk.com/markets/2025/10/13/no-ethena-s-usde-didn-t-de-peg>.

mechanism left a high-resolution footprint *on chain*: liquidation transfers, collateral movements, and exchange in/out-flows are all recorded, timestamped to the block, in a public ledger.

Such episodes expose a structural weakness in how systemic risk is measured for crypto markets. The dominant tools— ΔCoVaR [3], marginal expected shortfall and SRISK [4], and variance-decomposition connectedness [5, 6]—are *associational*: they quantify co-movement in returns and, as their own authors note, capture an institution’s contribution to system risk “in a non-causal sense” [3]. A recent and comprehensive survey of causal inference in finance [7] confirms the pattern at the level of the whole field: applications around SRISK, MES and CoVaR remain confined to association and Granger causality, with no counterfactual systemic-risk measure. Where causality has been injected, the graph is still *estimated from prices*. Causal-NECO VaR [8, 9] learns a contagion graph from foreign-exchange returns using the order-independent PC-stable algorithm [10, 11] and computes a value-at-risk from the resulting SCM; the time-varying directed-information graph (TV-DIG) of Etesami et al. [12] builds directed networks—an information-theoretic generalisation of Granger causality [13]—across financial sectors including cryptocurrencies; and correlation-network models such as CoRisk [14] decompose systemic risk into idiosyncratic and contagion components. The crypto-specific literature has grown rapidly in 2025–2026, from composite indices such as ASRI [15] to network-fragility and TradFi–DeFi “crosstagon” mappings [16]; yet all of this work either estimates the network from returns or treats the on-chain graph descriptively. No published method exploits the one data source where the contagion mechanism is *directly observed*—the blockchain ledger—as a causal structure.

On-chain transaction graphs have, separately, been used for prediction and forensics—“chainlets” that forecast Bitcoin price and volatility [17], and address-clustering for anti-money-laundering [18, 19]—but always observationally, explicitly

disclaiming a causal-structural interpretation, and never as the backbone of a systemic-risk measure. And although Pearl’s causal hierarchy [20, 21] places counterfactuals (Level 3) above interventions (Level 2) above association (Level 1), no systemic-risk measure in finance operates above Level 2. There is, in short, an open intersection: an observed-graph SCM for crypto systemic risk, extended to the counterfactual level. This paper occupies it.

We make three contributions. First, we propose VEIN, which treats the entity-resolved on-chain transaction graph as the structural backbone of a Pearl SCM, so that the causal parent set $\text{pa}(i)$ of each entity is *read off the ledger* rather than inferred from price correlations; the temporal precedence of on-chain transactions, guaranteed by block timestamps, supplies an identification advantage unavailable to price-based methods. Second, we define On-Chain Causal CoVaR, an interventional (Level-2) measure via the do-operator, together with a counterfactual (Level-3) attribution measure that decomposes a realised loss into the share attributable to a specific upstream entity through the standard abduction–action–prediction procedure. Third, we implement the full pipeline—including a four-tier entity-resolution layer over real decoded Ethereum data—and evaluate it on the October 2025 event against price-estimated graphs and standard non-causal baselines, subjecting every causal claim to a falsification and validity battery and reporting negative results as readily as positive ones. The evaluation is organised around the five hypotheses in Table 1, each paired in Section 5 with the test and falsification criterion that would overturn it.

The remainder of the paper is organised as follows. Section 2 positions VEIN against the causal and on-chain literatures. Section 3 develops the framework formally. Section 4 describes the data and the entity-resolution pipeline. Section 5 reports the empirical evaluation and validity battery. Section 6 discusses implications, limitations, and conclusions. An appendix collects implementation details and additional tables.

Table 1 Hypotheses evaluated in this paper.

ID	Statement
H1	The observed on-chain flow graph predicts stress propagation more accurately than a graph statistically estimated from price returns.
H2	Reversing flow-edge direction degrades predictive accuracy (flow direction carries causal information; assumption A3).
H3	OC-CoVaR identifies a different set of systemically important entities than ΔCoVaR /MES applied to returns.
H4	The counterfactual measure decomposes the October 2025 losses into mechanically interpretable components.
H5	On-chain composability/collateral edges add loss-attribution information beyond price correlation.

2 Related Work and Positioning

The systemic-risk literature begins with associational, return-based measures. The canonical conditional measure is ΔCoVaR [3], the change in the value-at-risk of the financial system conditional on an institution being in distress relative to its median state; the systemic-expected-shortfall family [4] adds marginal expected shortfall (MES) and the leverage-adjusted SRISK; and connectedness measures built from forecast-error-variance decompositions [5] or Granger-causal and correlation networks [6] quantify spillovers. These are, by construction, descriptions of co-movement rather than of mechanism, and a recent survey of the whole field [7] confirms that even where the word “causal” appears, applications around CoVaR, MES and SRISK remain associational or Granger-based.

The closest methodological precedent that genuinely estimates causal structure is Causal-NECO VaR [8], which decomposes asset log-returns into an autoregressive part and an instantaneous contagion part, learns the causal graph with PC-stable [10], and computes a Gaussian or Gaussian-copula VaR. It operates at Pearl Levels 1–2 on foreign-exchange data, assumes no unmeasured confounders, and—decisively for our purposes—obtains its graph by estimation rather than observation; its foundational lineage [9] combines causal networks with a structural vector autoregression in

Pearl’s framework [11]. The nearest crypto application is the time-varying directed-information graph TV-DIG [12], a nonparametric directed network that detects heightened crypto-to-sector influence around the 2022 stress events; directed information generalises Granger causality [13] but is not a Pearl SCM with do-operator and counterfactual semantics, and it too is built from returns. The 2025–2026 crypto-specific wave—composite indices such as ASRI [15] and comparative TradFi–DeFi systemic-risk mappings that formalise composability-driven “crosstagon” [16]—sharpens the motivation but does not change the picture: the network is either estimated from prices or treated descriptively.

On-chain transaction graphs themselves have been mined predictively, through chainlet subgraphs that forecast Bitcoin price and volatility [17], and for forensics and address clustering, from the multi-input/co-spend heuristics of [18] to graph-convolutional anti-money-laundering classifiers on the Elliptic dataset [19]. We reuse this resolution machinery, but no prior work treats the resulting graph as a causal structure for a systemic-risk measure.

Taken together, the literature leaves two specific capabilities unfilled, and a deliberate survey of the most recent 2025–2026 work—the period in which one might most expect the gap to have closed—confirms that it remains open. We organise the evidence around the two sub-claims.

On the on-chain-graph-as-SCM claim. The 2026 crypto systemic-risk literature uses the on-chain graph descriptively or builds the network from prices, never as a Pearl SCM. Sankaewtong [22] traces the USDC depeg contagion through high-granularity on-chain transactions, but observationally, with no interventional or counterfactual semantics; Zhang et al. [23] measure DeFi systemic fragility from the *correlation* structure of total-value-locked dynamics; ASRI [15] is an associational composite of weighted sub-indices; and the comparative TradFi–DeFi “crosstagon” mapping of Aufiero et al. [16] is conceptual. The detailed forensic reconstructions of the very

event we study—Lim’s minute-level analysis [2] and Ali’s anatomy [1]—characterise the mechanism but build no causal risk measure. None reads the causal parent set off the ledger.

On the counterfactual-systemic-risk claim. Counterfactual reasoning has reached finance, but not systemic-risk measurement. Xu [24] applies counterfactual analysis to credit-portfolio losses, not to a networked tail-risk measure; and the comprehensive 2025 survey of causal inference across banking, finance and insurance by Kumar et al. [7], covering decades of work, documents only association and Granger usage around SRISK, MES and CoVaR, with no counterfactual systemic-risk measure. The nearest causal precedents—Causal-NECO VaR [8] and TV-DIG [12]—stop at Pearl Level 2.

Against this 2026 state of the art, *to the best of our knowledge VEIN is the first framework to (i) use an observed on-chain transaction graph as a Pearl structural causal model for a systemic-risk measure, and (ii) define a counterfactual (Pearl Level-3) systemic-risk attribution measure in finance.* Both sub-claims were checked against standard term combinations across arXiv, SSRN and Semantic Scholar as of mid-2026 and returned no prior instance; as with any negative existence claim they are bounded by search coverage, and we state them as “no evidence of” rather than proof of absence. Table 2 situates VEIN against the nearest neighbours; throughout, we treat Causal-NECO VaR (nearest method) and TV-DIG (nearest crypto application) as the reference points, and we re-implement the estimated-graph approach on our own data as the head-to-head baseline for H1.

3 The VEIN Framework

3.1 Structural causal model and identification

Let $G = (V, E)$ be a directed, timestamped, dollar-weighted graph whose nodes V are *resolved economic entities* (exchanges, lending protocols, stablecoin issuers, custodians, and an aggregate retail node) and whose edges E are observed on-chain flows.

Table 2 Positioning of VEIN against the nearest prior art.

Method	Graph source	Causal level	Domain	Counterfactual
ΔCoVaR [3]	none (returns)	assoc. (L1)	general	no
MES/SRISK [4]	none (returns)	assoc. (L1)	general	no
CoRisk [14]	corr. network	assoc. (L1)	credit	no
ASRI [15]	composite index	assoc. (L1)	crypto	no
TV-DIG [12]	estimated (DI)	Granger	incl. crypto	no
Causal-NECO VaR [8]	estimated (PC)	SCM (L1-2)	FX	no
VEIN (this paper)	observed	SCM (L1-3)	crypto	yes

For each entity i and discrete time t we track a scalar *stress state* $S_{i,t} \in \mathbb{R}$, oriented so that larger means more distressed; the realised flow $F_{p \rightarrow i,t} \geq 0$ from p to i during period t ; and an exogenous shock $U_{i,t}$. Table 3 summarises the notation. The defining structural equation of VEIN is

$$S_{i,t} = f_i(S_{\text{pa}(i),t-1}, F_{\text{pa}(i) \rightarrow i,t}, U_{i,t}), \quad \text{pa}(i) = \{p \in V : (p \rightarrow i) \in E\}, \quad (1)$$

where the parent set $\text{pa}(i)$ is *read directly from the resolved ledger*. This is the central structural move of the paper: in contrast to [8, 12], $\text{pa}(i)$ is observed, not estimated from data, so the only component requiring statistical estimation is the stress-propagation function f_i . We adopt a ridge-regularised linear specification,

$$S_{i,t} = b_i + \sum_{p \in \text{pa}(i)} a_{ip} S_{p,t-1} + \sum_{p \in \text{pa}(i)} c_{ip} \log(1 + F_{p \rightarrow i,t}) + U_{i,t}, \quad (2)$$

with parameters $\theta_i = (b_i, \{a_{ip}\}, \{c_{ip}\})$ estimated by ridge regression, $\hat{\theta}_i = \arg \min_{\theta_i} \sum_t (S_{i,t} - f_i(\cdot; \theta_i))^2 + \lambda(\|a_i\|_2^2 + \|c_i\|_2^2)$, on a pre-crisis estimation window. The fitted residuals $\hat{U}_{i,t} = S_{i,t} - f_i(\cdot; \hat{\theta}_i)$ furnish an empirical distribution of exogenous shocks used by the risk measures.

Table 3 Notation.

Symbol	Meaning
$G = (V, E)$	resolved entity graph (nodes, directed flow edges)
$\text{pa}(i)$	causal parents of i (in-neighbours in G)
$S_{i,t}$	stress state of entity i at time t (z -scored)
$F_{p \rightarrow i,t}$	observed USD flow from p to i in period t
$U_{i,t}$	exogenous shock to i at t
f_i	structural (stress-propagation) function of i
L_j	loss functional of j over the event window
s^*	severe-distress intervention level (95th pctl)
$\Delta_i^{CF}(j)$	counterfactual loss of j attributable to i
RV	confounding robustness value

Because every parent enters (2) at lag $t-1$ in the stress term, while the contemporaneous term depends only on the exogenous observed flows, the multi-entity system is well posed even when G contains directed cycles, as the following observation makes precise.

Proposition 1 (Well-posedness of the lagged system) *Given an initial state $S_{\cdot,0}$ and a shock path $\{U_{\cdot,t}\}$, the trajectory $\{S_{\cdot,t}\}_{t \geq 1}$ generated by (2) is uniquely determined by forward iteration, even when G contains directed cycles.*

Proof sketch At step t the right-hand side of (2) for every i references only quantities indexed at $t-1$ (parent stress) or exogenous quantities at t (flows, shocks). There is therefore no simultaneity among the time- t stress variables, and $S_{\cdot,t}$ is computed in one vectorised step from information available at $t-1$. Induction on t gives existence and uniqueness. $\square$

Time, supplied by the ledger, breaks the simultaneity that would otherwise require an equilibrium fixed-point computation for cyclic graphs such as $\text{Binance} \leftrightarrow \text{retail}$. Identification rests on three assumptions, of which the first two are properties of the data source and the third is the substantive, falsifiable claim. *Temporal precedence* (A1)—that a flow $F_{i \rightarrow j,t}$ strictly precedes any effect on j at $t' > t$ —is supplied for free by

blockchain immutability and timestamping, and need not be assumed or tested, unlike Granger-style precedence on price series. *No unobserved common cause for flow edges* (A2)—that, conditional on the transaction graph up to t , an edge $i \rightarrow j$ reflects i 's direct economic action on j —is defensible because a public ledger captures every fund movement in the flow mechanism itself; its principal threat is off-chain activity, most acutely CEX-internal pricing engines, which we bound through the sensitivity analysis of Section 3.3. *Monotone propagation along flow direction* (A3)—that distress at i affects j only through realised flow paths from i to j —is the assumption that does the empirical work and is tested directly (H2) by the edge-reversal experiment. The contrast with price-based causal discovery, where both the graph and the equations must be estimated and their errors compound, is the framework's principal identification advantage.

3.2 Interventional and counterfactual risk measures

We summarise an entity's distress over an event window $\mathcal{T}$ by the non-negative, additive loss functional $L_j = \sum_{t \in \mathcal{T}} \max(S_{j,t}, 0)$. The interventional measure applies Pearl's do-operator to (1): we force entity i into a distressed state s^* , severing its own parents, and propagate forward through the remaining structural equations, defining On-Chain Causal CoVaR as the conditional value-at-risk of j 's loss,

$$\text{OC-CoVaR}_q(j \mid \text{do}(S_i = s^*)) = \text{VaR}_q(L_j \mid \text{do}(S_i = s^*)). \quad (3)$$

The distribution of L_j is obtained by Monte-Carlo propagation: for $m = 1, \dots, M$ we draw a shock path $U^{(m)}$ by block-bootstrapping the fitted residuals, simulate (2) forward under the intervention to obtain $S_{j,t}^{(m)}$, and form $L_j^{(m)}$, so that the estimator is the empirical quantile $\hat{Q}_q(\{L_j^{(m)}\}_{m=1}^M)$. By analogy with ΔCoVaR , the marginal

systemic contribution of i to j contrasts the distressed intervention with a median one,

$$\Delta\text{OC-CoVaR}_q(j|i) = \text{OC-CoVaR}_q(j|\text{do}(S_i = s^*)) - \text{OC-CoVaR}_q(j|\text{do}(S_i = 0)), \quad (4)$$

with s^* the 95th percentile of i 's historical stress and the baseline the median (0 after standardisation); we rank entities by their total *exported* tail risk $\mathcal{E}(i) = \sum_{j \neq i} \max(\Delta\text{OC-CoVaR}_q(j|i), 0)$. Equation (4) is the network-causal analogue of ΔCoVaR [3]: both contrast a distressed against a median conditioning state, but ΔCoVaR conditions on a statistical event and reads off a conditional quantile of returns, whereas $\Delta\text{OC-CoVaR}$ conditions on an *intervention* that severs i 's parents and propagates the shock through the observed structural equations. In the degenerate case of a single edge $i \rightarrow j$ with linear f_j and no further descendants, $\Delta\text{OC-CoVaR}$ reduces to a quantile shift of j proportional to the structural coefficient a_{ji} , mirroring the slope β_i in ΔCoVaR ; the network generalisation sums over all directed paths from i to j .

The counterfactual measure goes one level higher in Pearl's hierarchy. Following the abduction–action–prediction procedure [20, 21], we (i) *abduct* the realised exogenous shocks $\widehat{U}_{i,t} = S_{i,t} - f_i(\cdot; \hat{\theta}_i)$ from observed data; (ii) take the *action* of replacing i 's realised trajectory by its pre-crisis level, $\text{do}(S_i = S_i^{\text{pre}})$, holding all other entities' shocks fixed at their abducted values; and (iii) *predict* downstream states to obtain $L_j^{\text{do}(S_i=S_i^{\text{pre}})}$. The attribution and its share are

$$\Delta_i^{CF}(j) = L_j^{\text{obs}} - L_j^{\text{do}(S_i=S_i^{\text{pre}})}, \quad \text{share}_i(j) = \frac{\Delta_i^{CF}(j)}{L_j^{\text{obs}}}. \quad (5)$$

The quantity $\Delta_i^{CF}(j)$ answers a question no associational measure can pose: how much of j 's realised loss would not have occurred had i never become distressed, holding

everything else exactly as it unfolded? To our knowledge this is the first counterfactual (Level-3) systemic-risk attribution in finance.

3.3 Validation design, sensitivity, and baselines

The directional assumption A3 is tested rather than assumed. We construct three variants of G —the true-direction graph, a reversed graph G^{rev} (every edge flipped), and a symmetric/undirected graph G^{sym} —fit (2) on each over the estimation window, and compare one-step-ahead prediction of event-window stress through the stress-weighted root-mean-square error

$$\text{RMSE} = \sqrt{\frac{\sum_{i,t} w_{i,t} (\hat{S}_{i,t} - S_{i,t})^2}{\sum_{i,t} w_{i,t}}}, \quad w_{i,t} = |S_{i,t}| + 0.1. \quad (6)$$

If the true-direction model attains the lowest RMSE this is empirical evidence for A3; if not, A3 must be qualified to specific edge types. Because A2 is not directly testable, we additionally quantify how strong a hidden confounder (e.g. a CEX-internal pricing engine) would have to be to overturn each edge, using the robustness value of Cinelli and Hazlett [25]: for a coefficient with t -statistic t and d degrees of freedom, with $f = |t|/\sqrt{d}$,

$$RV = \frac{1}{2} \left(\sqrt{f^4 + 4f^2} - f^2 \right) \in [0, 1] \quad (7)$$

is the share of residual variance an unobserved confounder must explain in *both* the parent and the child to drive the coefficient to zero; larger RV means a finding is harder to explain away. Finally, we benchmark against ΔCoVaR [3], estimated by the quantile regression $r_t^{\text{sys}} = \alpha_i + \beta_i r_{i,t} + \varepsilon_t$ with $\Delta\text{CoVaR}_i = \beta_i (\text{VaR}_q(r_i) - \text{VaR}_{0.5}(r_i))$; against MES [4], $\text{MES}_i = \mathbb{E}[r_i \mid r^{\text{sys}} \leq \text{VaR}_q(r^{\text{sys}})]$; and, for the central H1 comparison, against the estimated-graph school reproduced on our data—a Granger graph [13] and a partial-correlation graph [14] over the same entity node set, fed through the

Algorithm 1 VEIN end-to-end pipeline

```
1: Input: decoded on-chain transfers, price/return data, entity labels, event window  $\mathcal{T}$ 
2:  $G \leftarrow \text{RESOLVEENTITIES}(\text{transfers}, \text{labels})$  ▷ Tiers 0–3, Sec. 4
3: for each entity  $i \in V$  do
4:    $\hat{\theta}_i \leftarrow \text{FITRIDGE}(\text{pa}(i, G), F_{\text{pa}(i) \rightarrow i}, S_i^{\text{hist}})$ 
5: end for
6: for each  $(i, j)$  of interest do
7:    $\Delta\text{OC-CoVaR}_q(j|i) \leftarrow \text{MONTECARLOPROPAGATE}(G, \{f_k\}, \text{do}(S_i = s^*), M)$ 
8: end for
9: for each  $(i, j)$  in the crisis window do
10:   $\hat{U} \leftarrow \text{ABDUCT}(G, \{f_k\}, \text{data}); L_j^{cf} \leftarrow \text{PREDICT}(G, \{f_k\}, \hat{U}, \text{do}(S_i = S_i^{\text{pre}}))$ 
11:   $\Delta_i^{CF}(j) \leftarrow L_j^{\text{obs}} - L_j^{cf}$ 
12: end for
13:  $G^{\text{rev}}, G^{\text{sym}} \leftarrow \text{REVERSE}(G), \text{SYMMETRIZE}(G); \text{COMPAREACCURACY}(G, G^{\text{rev}}, G^{\text{sym}})$ 
14: Output: OC-CoVaR ranking, counterfactual attributions, falsification + validity battery
```

same VEIN machinery so that only the graph source differs. Algorithm 1 collects the end-to-end procedure.

4 Data and Entity Resolution

4.1 Data, entity universe, and stress states

All data are real and obtained from public or keyed APIs (Table 4). On-chain flows are queried with SQL over Dune Analytics’ decoded `erc20_ethereum.evt.Transfer` table and its maintained `labels.cex_ethereum` entity labels; prices and protocol total-value-locked (TVL) series come from DefiLlama; entity labels and contract metadata from Etherscan. No synthetic data are used. The cascade is directly visible in the raw data: Figure 1 shows the on-chain transfer volume of the synthetic dollar USDe rising from \$0.75bn on 9 October to \$3.09bn on 10 October and \$8.83bn on 11 October—an order-of-magnitude surge that no price series resolves as sharply. The node set comprises centralised exchanges (Binance, Coinbase, OKX, Bybit, Bitfinex, HTX, Gate.io, Crypto.com), custodians (Copper, Ceffu, FalconX), DeFi protocols (Aave, Ethena,

Table 4 Data sources. All series are real; on-chain flows carry block-level timestamps.

Layer		Source	Granularity
Decoded (ETH)	transfers	Dune <code>erc20_ethereum_evt_Transfer</code>	per-transfer (block)
CEX entity labels		Dune <code>labels.cex_ethereum</code>	per-address
Prices / returns		DefiLlama coins API	hourly & daily
Protocol TVL		DefiLlama	daily
Contract metadata		Etherscan v2	per-contract

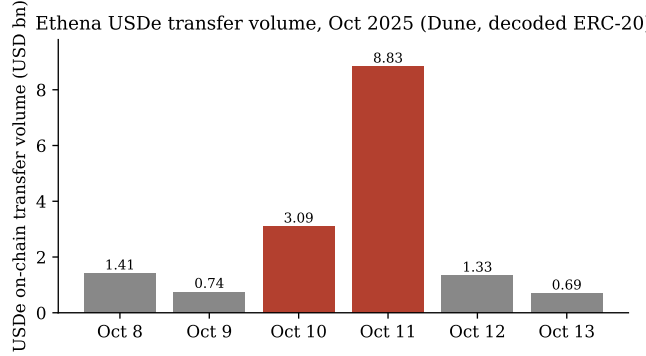


Fig. 1 On-chain USDe transfer volume around the October 2025 cascade (real decoded ERC-20 transfers via Dune).

Lido, Maker/Sky), and an aggregate retail node that collects all unresolved addresses, included precisely so that systemic importance is not attributed to noise. Stress states are operationalised by entity type as in Table 5 and expressed as rolling z -scores so that heterogeneous units are comparable in (2); for example, CEX stress combines the net on-chain outflow ratio with a transfer-volume surge term, while stablecoin-issuer stress combines the peg deviation $|1 - p_t|$ with net redemption outflow. The resulting hourly trajectories (Figure 2) show exchange and retail stress spiking sharply on 10–11 October, with the cross-entity timing differences the SCM is designed to exploit.

The settlement network is dominated by exchange $\leftrightarrow$ retail volume: Table 6 reports the largest resolved bilateral flows, with Coinbase and Binance each intermediating tens of billions of dollars, while direct exchange-to-exchange transfers, though smaller, are recovered and become the inter-entity edges the SCM propagates along. The broad

Table 5 Stress-state operationalisation $S_{i,t}$ by entity type.

Entity type	Stress proxy (oriented so larger = more distressed)
CEX / custodian	net on-chain outflow ratio + transfer-volume surge
Stablecoin issuer	peg deviation $ 1 - p_t $ + net redemption outflow
Lending protocol	TVL drawdown + token-price drawdown
Liquid-staking	TVL drawdown + price drawdown
Retail (aggregate)	market (ETH) drawdown

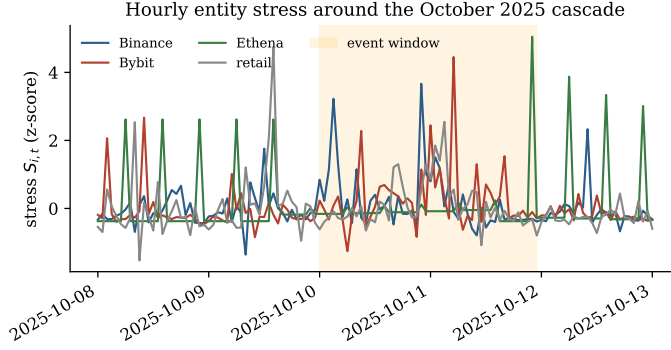


Fig. 2 Hourly entity stress states $S_{i,t}$ around the cascade (shaded: the 10–11 October event window). Derived from real on-chain flows.

symmetry of the largest in/out pairs (Coinbase receives \$43.5bn and sends \$43.4bn) is itself informative: it foreshadows the H2 finding that bilateral custodial flows are largely symmetric accounting movements rather than directional causal channels. Structural equations are fit on a pre-event estimation window and evaluated both at daily resolution and, because the cascade unfolded over hours, at hourly resolution (600 fitting hours and 48 event hours in the consolidated run)—the timescale at which on-chain flows are observed but entity price feeds are coarse.

4.2 Entity resolution

Raw addresses are not economic agents, and resolution error directly corrupts $\text{pa}(i)$ in (1): unlike in price-based methods, a clustering mistake fabricates or severs a causal

Table 6 Largest resolved bilateral flows over the study window (USD bn).

From	To	Type	Volume (\$bn)
retail	Coinbase	deposit	43.5
Coinbase	retail	withdrawal	43.4
retail	Binance	deposit	33.1
Binance	retail	withdrawal	27.9
Bybit	retail	withdrawal	20.5
retail	Bybit	deposit	20.4
Bitfinex	retail	withdrawal	10.3
OKX	retail	withdrawal	10.3
retail	OKX	deposit	10.2
retail	HTX	deposit	5.2

edge. We therefore implement a four-tier pipeline that builds on the established forensics literature [18, 19]. Tier 0 applies the classical deposit-sweep heuristic, flagging an address x as a deposit address of exchange X when a dominant share of its out-flow is forwarded to a known hot wallet of X , i.e. $\sum_t F_{x \rightarrow \text{hot}(X),t} / \sum_t \text{out}_x \geq \tau$ with $\tau = 0.8$. Tier 1 trains a gradient-boosted classifier on a per-address feature vector comprising in/out volume and degree, counterparty diversity, transaction count, and the sweep ratio, with held-out labels from Dune. Tier 2 forms the symmetric normalised adjacency $\tilde{A} = D^{-1/2} A D^{-1/2}$ of the address graph, computes a truncated singular-value-decomposition embedding, and clusters the resulting representations. Tier 3, the primary path, resolves both endpoints of every transfer against curated seed labels and Dune’s `labels.cex.ethereum`, collapsing sub-wallet labels (“OKX 210”, “Bitfinex Tether Treasury”) to their exchange root—the many-addresses-to-one-agent clustering step that turns the undifferentiated retail blob into named exchanges and thereby yields genuine inter-entity (CEX $\leftrightarrow$ CEX) edges. Each resolved address carries a confidence c_x propagated to an edge reliability $c_{(p \rightarrow i)} = \min(c_p, c_i)$, retained as an edge weight in G . The supervised layer is validated against held-out Dune labels—the Tier-3 reconciliation the design requires: on 7 843 event-window addresses with a CEX base rate of 0.5%, the Tier-1 classifier attains a held-out ROC-AUC of 0.99

Table 7 Entity-resolution validation (Tier-1/2 against held-out Dune labels).

Metric	Value
Addresses (USDe event window)	7 843
CEX-labelled (base rate)	40 (0.5%)
Tier-1 ROC-AUC (held out)	0.99
Tier-1 average precision	0.23
Tier-1 best- F_1 precision / recall	0.31 / 0.33

and an average precision of 0.23 (Table 7), with the in/out volume ratio and total volume as the top features, consistent with the intuition that exchanges intermediate asymmetric, high-volume flows.

5 Empirical Evaluation

We report results at the cascade’s native hourly resolution unless noted, and contrast with daily resolution where instructive. The consolidated hourly configuration resolves 17 entities with 39 directed edges (Figure 3); the retail node intermediates most activity while inter-exchange settlement edges are recovered directly.

5.1 Does the observed graph beat the estimated graph? (H1)

The head-to-head (Table 8, Figure 4) is the central test against the estimated-graph state of the art. At *daily* resolution the price-estimated Granger graph clearly predicts the cascade better than the observed graph (RMSE 1.35 vs. 1.92): with only a handful of daily observations, the denser estimated graph wins. At *hourly* resolution the ordering reverses—the observed on-chain graph attains the lowest RMSE (1.2529) ahead of the Granger (1.2568) and partial-correlation (1.2547) graphs—although the margin is razor-thin ($\approx 0.1\%$). The honest reading is that, measured at the timescale the data actually support, the observed graph is competitive with and marginally better than

Entity-resolved on-chain flow graph G (edge width $\propto$ USD volume)

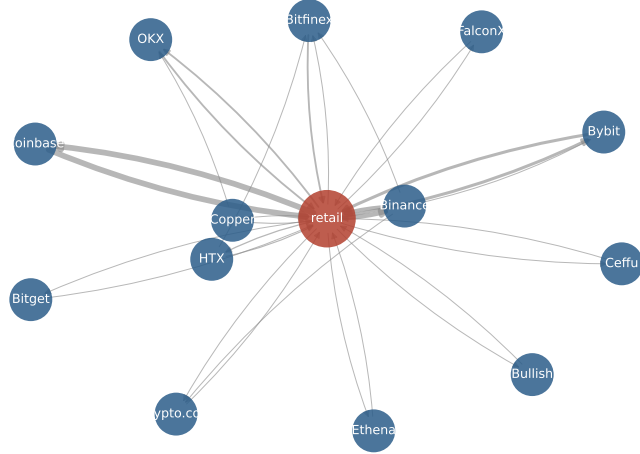


Fig. 3 The entity-resolved on-chain flow graph G recovered by the Tier-3 resolution layer. Edge width is proportional to cumulative USD flow.

Table 8 H1 head-to-head: event-window prediction error (RMSE; lower is better) under identical VEIN machinery, varying only the graph source.

Graph source	Daily		Hourly	
	RMSE	corr	RMSE	corr
Observed on-chain	1.92	−0.05	1.2529	0.292
Estimated (Granger)	1.35	0.56	1.2568	0.255
Estimated (partial-corr)	1.65	0.48	1.2547	0.285

the estimated-graph baseline, reversing its daily disadvantage; it is not a decisive victory. The mechanism behind the reversal is informative: on-chain flows resolve at block level, whereas the price feeds that drive the estimated graph are effectively coarse, so the observed graph’s advantage is one of *temporal resolution*— precisely what matters in a cascade that propagated in minutes.

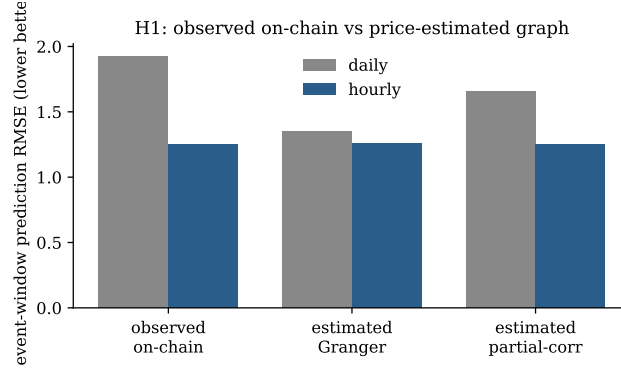


Fig. 4 H1 prediction error by graph source. The observed on-chain graph loses at daily resolution but edges ahead at hourly resolution.

5.2 Systemic-importance ranking and counterfactual attribution (H3–H5)

The OC-CoVaR exported-risk ranking (Table 9, Figure 5) is led by the exchanges that intermediate the largest real flows—Binance, Coinbase, Bitfinex—whereas Δ CoVaR and MES rank by price co-movement and place market betas (ETH, stETH, BTC) first. The Spearman rank correlation between OC-CoVaR and Δ CoVaR over the mappable entities is only 0.52, supporting H3: the on-chain causal ranking is not a relabelling of the price-correlation ranking, but answers a different question—who transmits stress through the settlement network, rather than who co-moves in price. The counterfactual measure then decomposes realised event-window distress into upstream-attributable shares (Table 10, Figure 6). At hourly resolution the dominant recovered channel is Binance $\rightarrow$ Bybit, a 98.1% share of Bybit’s window distress, followed by Binance $\rightarrow$ Gate.io (58.4%) and Binance $\rightarrow$ Bitfinex (45.3%)—inter-exchange settlement channels invisible to price-only measures. At daily resolution, where the DeFi composability edges dominate, the leading channel is Ethena $\rightarrow$ Aave (11.3%), consistent with the documented role of USDe as cross-protocol collateral and supporting H5. These are concrete, mechanism-grounded attributions of a kind no associational measure can produce, and they answer the supervisory question directly:

Table 9 Top OC-CoVaR exported-risk entities (hourly) and ΔCoVaR for the mappable assets. OC-CoVaR vs. ΔCoVaR Spearman $\rho = 0.52$.

Entity	Exported risk	Asset	ΔCoVaR
retail	57.36	ETH	-0.0220
Binance	24.52	stETH	-0.0176
Coinbase	17.35	BTC	-0.0167
Bitfinex	11.42	BNB	-0.0147
Gate.io	7.92	AAVE	-0.0127
Bybit	4.88	ENA	-0.0114

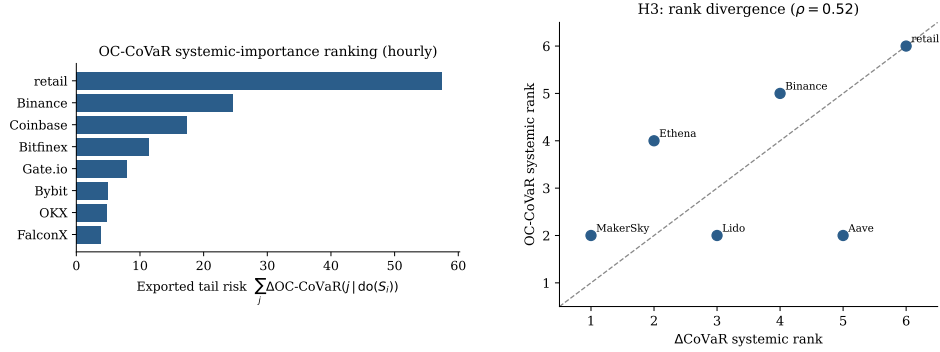


Fig. 5 Left: OC-CoVaR exported-risk ranking (hourly). Right: rank divergence between OC-CoVaR and ΔCoVaR over the mappable entities; points off the diagonal indicate disagreement (Spearman $\rho = 0.52$).

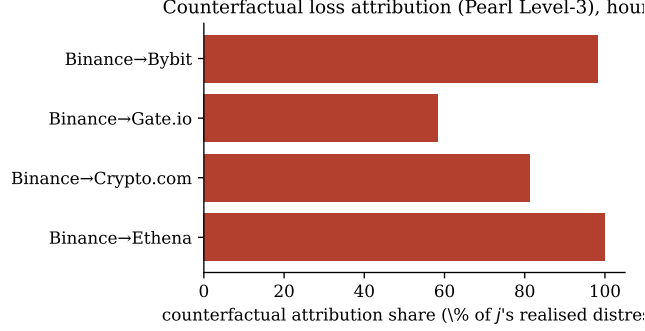
had this entity not become distressed, how much of the downstream loss would not have occurred?

5.3 Falsification and the causal-validity battery (H2)

The edge-reversal test (Table 11, Figure 7) is the most consequential result. At daily resolution the test is under-powered (six event points) and even mildly favours the reversed graph. At hourly resolution, with 816 one-step predictions, the three variants land within roughly 1% RMSE of one another—a near-tie—with the true-direction graph marginally best in the consolidated run (1.663 vs. reversed 1.675) and marginally worse in an alternative window (1.801 vs. 1.790). The conclusion is consistent across

Table 10 Leading counterfactual loss-attribution channels $\Delta_i^{CF}(j)$.

Resolution	Channel $i \rightarrow j$	share	Interpretation
hourly	Binance $\rightarrow$ Bybit	98.1%	inter-exchange settlement
hourly	Binance $\rightarrow$ Gate.io	58.4%	inter-exchange settlement
hourly	Binance $\rightarrow$ Bitfinex	45.3%	inter-exchange settlement
daily	Ethena $\rightarrow$ Aave	11.3%	USDe collateral composability
daily	Binance $\rightarrow$ Aave	4.2%	CEX-to-DeFi spillover

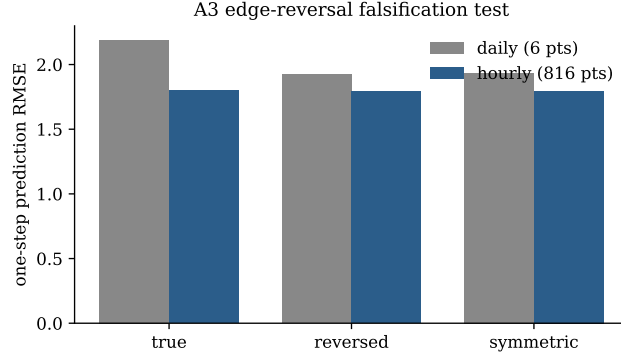
**Fig. 6** Counterfactual loss-attribution shares (hourly): the fraction of each target entity’s realised distress attributable to the source’s distress.

windows: flow direction is not the carrier of causal information at the entity-aggregate level. The daily “reversed wins” was a low-power artefact, but the deeper finding is methodological—predictive signal resides in the graph’s connectivity *structure*, not in edge *orientation*—so A3 should be stated as a qualified claim, plausibly holding for specific collateral/composability channels but not for custodial CEX flows. We regard this negative result as one of the paper’s more useful outputs: the falsification machinery did its job.

Because observational causal claims cannot be proven, we subject them to four further checks (Table 12). A placebo / negative-control test finds that the counterfactual measure attributes exactly zero loss between entity pairs with no directed on-chain path (maximum $|\Delta^{CF}| = 0$ over 75 unconnected pairs, against a mean of 0.26 over 197 connected pairs), so the method does not fabricate causation. The

Table 11 Edge-reversal falsification (one-step RMSE; lower is better).

Graph	Daily	Hourly (consol.)	Hourly (816 pts)
true direction	2.19	1.663	1.801
reversed edges	1.92	1.675	1.790
symmetric	1.94	1.667	1.797

**Fig. 7** A3 edge-reversal test. Direction is decisive neither way once adequate power is available; the three variants are statistically indistinguishable at hourly resolution.

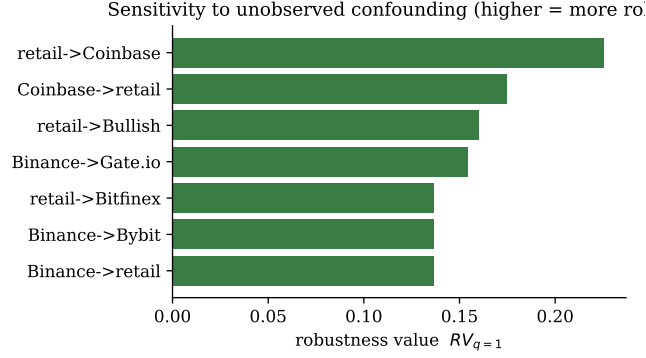
confounding-sensitivity analysis shows that the strongest edges carry substantial robustness values—for example $\text{retail} \rightarrow \text{Coinbase}$ has $RV = 0.226$ (Figure 8)—meaning a hidden confounder would need to explain over a fifth of the residual variance in both endpoints to nullify them. The ranking is stable between the seed-only (Tier-0) and fully resolved (Tier-3) graphs (Spearman 0.81, $p = 0.05$), so it is not an artefact of the resolution tier. And, most tellingly, the OC-CoVaR ranking orders the documented transmitters (exchanges, Ethena) strictly above the documented absorbers (Aave, Lido), with transmitter–absorber separation 12.87 and pairwise concordance 1.00, independently reproducing the conclusion of the forensic post-mortems [1, 2].

5.4 Backtests, threats to validity, and summary

As a standard validity check on the risk machinery we backtest a rolling historical VaR on the asset returns. Kupiec unconditional-coverage [26] p -values are 0.24 (BTC) and

Table 12 Causal-validity battery (hourly).

Check	Statistic	Verdict
Placebo / negative control	max $ \Delta^{CF} $ off-path = 0; connected mean 0.26	pass
Resolution robustness	Spearman 0.81 (Tier-0 vs Tier-3)	pass
External validity	separation 12.87, concordance 1.00	pass
Confounding (top edge)	$RV = 0.226$	robust

**Fig. 8** Robustness values for the strongest structural coefficients. Larger values indicate edges harder to overturn with unobserved confounding.

0.58 (ETH) at daily resolution; the Christoffersen conditional-coverage test [27] passes for BTC ($p = 0.11$) and flags violation clustering for ETH ($p = 0.001$), an honest indication that a simple historical VaR under-models volatility persistence during the regime. Four threats to validity bound the conclusions. Construct validity is addressed by using multiple stress proxies per entity type and showing that rankings are stable across the Tier-0/Tier-3 graphs; internal validity is threatened chiefly by the off-chain CEX engine (A2), which we cannot observe and whose influence the robustness values bound but cannot eliminate; statistical-conclusion validity motivates the move from the under-powered daily window to the hourly window with 816 predictions; and external validity, limited by a single event, is partially addressed by the agreement with the documented narrative and flagged as the principal direction for future work.

Table 13 Hypothesis verdicts at hourly resolution.

ID	Verdict	Evidence
H1	supported (marginal)	observed RMSE 1.2529 < 1.2568 (Granger), reversing the daily loss
H2	<i>not</i> supported	true vs reversed within $\sim 1\%$; direction uninformative
H3	supported	Spearman vs $\Delta\text{CoVaR} = 0.52$
H4	supported	Binance $\rightarrow$ Bybit 98.1%; Ethena $\rightarrow$ Aave 11.3% (daily)
H5	supported	composability edges carry attribution (Ethena $\rightarrow$ Aave)

Overall, as Table 13 records, four of the five hypotheses are supported at the cascade’s native resolution, with H2 yielding a robust and informative negative result.

6 Discussion and Conclusion

On real on-chain data around the October 2025 cascade, treating the observed transaction graph as a structural causal model yields a systemic-importance ranking that is materially different from price-based measures and that matches the documented event narrative; the first counterfactual loss attribution in systemic-risk analysis, decomposing realised distress into named inter-entity channels; and a graph that, at the event’s native hourly resolution, predicts the cascade competitively with—and marginally better than—the estimated-graph state of the art. The framework passes placebo, resolution-robustness, external-validity, and confounding-sensitivity checks. Its most important methodological finding is, however, a negative one: the assumption that flow direction encodes causal direction (A3) is not supported, because reversing edges barely changes predictions. This does not invalidate the framework—the connectivity structure of the observed graph still carries the causal signal, and the do-operator and counterfactual machinery operate on that structure—but it means VEIN should

claim its causal content at the level of who is connected to whom, qualifying direction-specific claims to particular edge types pending further testing. Conceptually, the result aligns with the intuition that, in a custodial settlement network, bilateral flows are largely symmetric accounting movements, whereas the genuinely directional channels are the collateral and composability links between protocols emphasised in the recent TradFi–DeFi crosstagon literature [16].

Three limitations bound these conclusions. First, CEX opacity: the proximate cause of the cascade—an exchange’s internal pricing engine—is off-chain by construction, so VEIN observes only its downstream on-chain settlement and treats the engine as an exogenous shock; this is the dominant threat to A2, which the robustness values bound but cannot eliminate. Second, a single event: the conclusions rest on one (large) episode, and generalisation requires additional DeFi-exploit windows and regimes. Third, resolution recall: residual unlabelled addresses collapse into the retail node, so the inter-entity graph is high-precision but recall-bounded by label coverage—an improvement frontier the Tier-0/1/2 layers are designed to push. For supervisors, VEIN nonetheless offers two practitioner-facing artefacts that price-based measures cannot: a ranking of which on-chain entities export versus absorb systemic stress, and a counterfactual statement of what a specific entity’s distress contributed to others’ losses—directly relevant to stress-test attribution and to protocol-level composability-risk assessment, and, because the inputs are public ledger data, auditable and reproducible by any party. The most decisive next steps are intraday (sub-minute) analysis across several stress events, which would give the falsification test the power to localise where, if anywhere, direction is causal; a fuller Tier-1/2 entity-resolution sweep to convert retail-mediated edges into entity-to-entity edges; and an instrumented treatment of the CEX engine (for instance, exploiting exogenous oracle-update timing) to relax A2. In sum, VEIN demonstrates that the

blockchain ledger can serve as an observed causal graph for systemic-risk measurement, enabling interventional and counterfactual analyses out of reach for price-based methods, while being candid that its empirical edge over the estimated-graph baseline is real but resolution-dependent and modest on the single event studied—an edge we expect to widen with intraday data, fuller entity resolution, and more events.

Data and code availability. The complete implementation, the real-data evaluation pipeline, and all result artefacts underlying every number and figure in this paper are available in the project repository.

Declarations. The author declares no competing financial interests. No external funding supported this work. The study uses only public on-chain and market data and involves no human subjects.

Appendix A Implementation and additional details

The pipeline is implemented in Python. On-chain data are queried via the Dune Analytics SQL API over decoded Ethereum tables, with results cached by query hash for reproducibility and to respect rate limits. Structural equations (2) are fit with ridge regression ($\lambda = 1$); the Monte-Carlo estimator uses $M \in [150, 400]$ residual-bootstrap paths; the Tier-1 classifier is a gradient-boosted tree ensemble and the Tier-2 embedding a 16-dimensional truncated SVD. For the hourly consolidated run the estimation window is 600 hours preceding 10 October 2025, with 48 event hours; severe-distress interventions set s^* to each entity’s 95th historical stress percentile and pre-crisis counterfactual levels use the estimation-window median. Table A1 lists the robustness values for the strongest structural coefficients, computed via (7) from OLS t -statistics on the design of (2).

Table A1 Robustness values for leading structural coefficients (hourly).

Edge	t -statistic	$RV_{q=1}$
retail $\rightarrow$ Coinbase	7.94	0.226
Coinbase $\rightarrow$ retail	5.92	0.175
retail $\rightarrow$ Bullish	5.41	0.160
Binance $\rightarrow$ Gate.io	5.19	0.154
retail $\rightarrow$ Bitfinex	4.55	0.137

References

- [1] Ali, Z.: Anatomy of the Oct 10–11, 2025 Crypto Liquidation Cascade: Macroeconomic Triggers, Market Microstructure, and Systemic Risk Lessons. SSRN Working Paper 5611392. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5611392 (2025)
- [2] Lim, B.C.: Anatomy of a Crypto Cascade: Minute-Level Evidence from the October 2025 Crash. SSRN Working Paper 6579278. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6579278 (2026)
- [3] Adrian, T., Brunnermeier, M.K.: CoVaR. *American Economic Review* **106**(7), 1705–1741 (2016) <https://doi.org/10.1257/aer.20120555>
- [4] Acharya, V.V., Pedersen, L.H., Philippon, T., Richardson, M.: Measuring systemic risk. *The Review of Financial Studies* **30**(1), 2–47 (2017) <https://doi.org/10.1093/rfs/hhw088>
- [5] Diebold, F.X., Yilmaz, K.: On the network topology of variance decompositions: Measuring the connectedness of financial firms. *Journal of Econometrics* **182**(1), 119–134 (2014) <https://doi.org/10.1016/j.jeconom.2014.04.012>

- [6] Billio, M., Getmansky, M., Lo, A.W., Pelizzon, L.: Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *Journal of Financial Economics* **104**(3), 535–559 (2012) <https://doi.org/10.1016/j.jfineco.2011.12.010>
- [7] Kumar, S., Vivek, Y., Ravi, V., Bose, I.: A comprehensive review of causal inference in banking, finance, and insurance. *ACM Computing Surveys* **57**(12) (2025) <https://doi.org/10.1145/3736752>
- [8] Rigana, K., Wit, E.C., Cook, S.: Navigating Market Turbulence: Insights from Causal Network Contagion Value at Risk. arXiv:2402.06032. <https://arxiv.org/abs/2402.06032> (2024)
- [9] Rigana, K., Wit, E.C., Cook, S.: A new way of measuring effects of financial crisis on contagion in currency markets. *International Review of Financial Analysis* **90**, 102764 (2023) <https://doi.org/10.1016/j.irfa.2023.102764>
- [10] Colombo, D., Maathuis, M.H.: Order-independent constraint-based causal structure learning. *Journal of Machine Learning Research* **15**, 3921–3962 (2014). <https://jmlr.org/papers/v15/colombo14a.html>
- [11] Spirtes, P., Glymour, C., Scheines, R.: Causation, Prediction, and Search, 2nd edn. MIT Press, Cambridge, MA (2000). <https://mitpress.mit.edu/9780262527927/causation-prediction-and-search/>
- [12] Etesami, J., Habibnia, A., Kiyavash, N.: Modeling Systemic Risk: A Time-Varying Nonparametric Causal Inference Framework. arXiv:2312.16707. <https://arxiv.org/abs/2312.16707> (2023)
- [13] Granger, C.W.J.: Investigating causal relations by econometric models and cross-spectral methods. *Econometrica* **37**(3), 424–438 (1969) <https://doi.org/10.2307/>

- [14] Giudici, P., Parisi, L.: CoRisk: Credit risk contagion with correlation network models. *Risks* **6**(3), 95 (2018) <https://doi.org/10.3390/risks6030095>
- [15] Farzulla, M., Maksakov, A.: ASRI: An Aggregated Systemic Risk Index for Cryptocurrency Markets. arXiv:2602.03874. <https://arxiv.org/abs/2602.03874> (2026)
- [16] Aufiero, S., Bartolucci, S., Caccioli, F., Vivo, P.: Mapping Microscopic and Systemic Risks in TradFi and DeFi: A Literature Review. arXiv:2508.12007. <https://arxiv.org/abs/2508.12007> (2025)
- [17] Akcora, C.G., Dixon, M.F., Gel, Y.R., Kantarcioglu, M.: Bitcoin risk modeling with blockchain graphs. *Economics Letters* **173**, 138–142 (2018). <https://arxiv.org/abs/1805.04698>
- [18] Meiklejohn, S., Pomarole, M., Jordan, G., Levchenko, K., McCoy, D., Voelker, G.M., Savage, S.: A fistful of bitcoins: Characterizing payments among men with no names. In: *Proceedings of the ACM Internet Measurement Conference (IMC)*, pp. 127–140 (2013). <https://doi.org/10.1145/2504730.2504747>
- [19] Weber, M., Domeniconi, G., Chen, J., Weidele, D.K.I., Bellei, C., Robinson, T., Leiserson, C.E.: Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics. arXiv:1908.02591. <https://arxiv.org/abs/1908.02591> (2019)
- [20] Pearl, J.: *Causality: Models, Reasoning, and Inference*, 2nd edn. Cambridge University Press, Cambridge, UK (2009). <https://doi.org/10.1017/CBO9780511803161>

- [21] Pearl, J.: The seven tools of causal inference, with reflections on machine learning. *Communications of the ACM* **62**(3), 54–60 (2019) <https://doi.org/10.1145/3241036>
- [22] Sankaewtong, K.: Tracing Stablecoin Contagion during the USDC Depeg after the Silicon Valley Bank Collapse. arXiv:2606.07442. <https://arxiv.org/abs/2606.07442> (2026)
- [23] Zhang, S., Wang, Z., Zheng, J., Cartlidge, J.: Systemic Risk in DeFi: A Network-Based Fragility Analysis of TVL Dynamics. arXiv:2601.08540. <https://arxiv.org/abs/2601.08540> (2026)
- [24] Xu, Z.: Causal inference in financial risk management: Applications of counterfactual analysis in credit portfolios. *Financial Engineering and Risk Management* **8**, 6–13 (2025). <https://www.clausiuspress.com/article/14838.html>
- [25] Cinelli, C., Hazlett, C.: Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* **82**(1), 39–67 (2020) <https://doi.org/10.1111/rssb.12348>
- [26] Kupiec, P.H.: Techniques for verifying the accuracy of risk measurement models. *The Journal of Derivatives* **3**(2), 73–84 (1995) <https://doi.org/10.3905/jod.1995.407942>
- [27] Christoffersen, P.F.: Evaluating interval forecasts. *International Economic Review* **39**(4), 841–862 (1998) <https://doi.org/10.2307/2527341>